



CSE340

Computer Architecture

Research-Based Assessment

Submitted by
Abdullah Al Fahad
ID: 21201240
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Submitted to - Partha Bhoumik (PBK)

Introduction

The PlayStation 5 (PS5) is a next generation gaming console that pairs a high bandwidth unified memory pool with a custom NVMe SSD and dedicated decompression hardware to deliver fast, low-latency game streaming and rendering (Patterson & Hennessy, 2020). PS5 organizes its memory to affect responsiveness, load times, and frame stability in a way that is very important for its performance. This report analyzes the PS5's cache, DRAM, and storage layers and explains how these memory choices help the console perform well while also showing some trade-offs (Strickland, 2020; Copetti, n.d.).

Background

Computer systems arrange the memory in layers to find a balance between speed, size, cost, and energy use is known as a hierarchy. The fastest but smallest parts, such as registers and caches, are placed near the processor; on the other hand, the bigger and slower parts, such as DRAM and SSD's are used for main memory and long-term storage. The success of this hierarchy relies on ideas like temporal and spatial locality, cache associativity, and average memory access time (Patterson & Hennessy, 2020). If we sum up the explanations of memory hierarchy, it is shown that higher-level caches reduce the cost of cache misses and lower the pressure on DRAM, while virtual memory gives both abstraction and protection (GeeksforGeeks, n.d.). These basic ideas strongly affect how real-world system design their memory level to achieve high responsiveness and efficiency.

The PlayStation 5 (PS5) already shows how these principles can be adapted to high-performance, cost-constrained hardware. Sony's previous gen consoles are relied on HDD's but the PS5 integrates a custom NVMe SSD that is capable of 5.5 GB per second raw throughput, paired with hardware decompression to stream assets on demand instead of preloading them into DRAM (Strickland, 2020). Moreover, it is an AMD-based SoC that employs a unified 16 GB GDDR6 memory pool, shared by both CPU and GPU. It also reduces redundant transfers but introduces potential contention during bandwidth-heavy workloads (Copetti, n.d.). The article on the PS5 Pro shows that Sony is scaling GPU throughput and memory bandwidth to address increasing fidelity demands, highlighting bandwidth as a key limiter (The Verge, 2024). However, the analyses are primarily focused on the performance benefits of SSD streaming and unified memory, but there is a gap in attention to long-term scalability with a fixed 16 GB pool, cache coherence overheads, or energy trade-offs under sustained workloads. This gap shows the need deeper examination of how the PS5's memory hierarchy balances performance, cost, and efficiency, and where improvements or alternative designs might be necessary.

Methodology

The articles examine the PS5 memory hierarchy by analyzing its Zen 2 CPU, RDNA 2 GPU, 16 GB GDDR6 memory pool, and custom NVMe SSD with hardware decompression, proposing optimizations based on established design principles. The analysis assumes publicly available specs are accurate since Sony's proprietary details are not disclosed. The PS5 benefits from a unified memory pool that simplifies development, along with a 5.5 GB/s SSD for real-time asset streaming (Strickland, 2020; Copetti, n.d.). However, the proposed redesign includes three improvements: (1) a hybrid model with a dedicated cache slice for CPU tasks, reducing contention with GPU workloads, (2) direct GPU-to-SSD communication through DMA to lower asset streaming latency by bypassing CPU scheduling, and (3) per-core power gating for better energy efficiency without sacrificing responsiveness. Theoretically, the system is a data flow between the CPU, GPU, and GDDR6 memory pool, where SSD acts as a memory extension to feed data through decompression hardware into RAM. These optimizations will look to reduce load times, stabilize multitasking performance, lower energy usage per operation, and keep system costs manageable through design modification rather than expensive hardware upgrades.

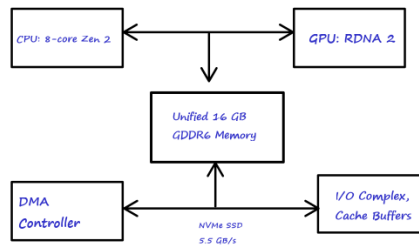


Fig: Diagram

Discussion

The proposed improvements to the PS5 memory hierarchy would significantly boost gaming performance and responsiveness. By incorporating a hybrid memory model with a CPU-reserved cache slice, CPU tasks like AI and background processes would experience fewer slowdowns during GPU rendering peaks, reducing frame drops and multitasking delays. Direct GPU-to-SSD communication would further lower latency in streaming high-resolution assets, enhancing load times and open-world rendering fluidity. Meanwhile, fine-grained per-core power gating would improve energy efficiency, reducing thermal output without compromising performance. However, these methods come with trade offs such as the hybrid memory model adds complexity and may decrease overall GPU bandwidth more,over, SSD,GPU DMA transfers require additional scheduling, which could introduce bottlenecks. This underscores the need for effective software hardware coordination. Techniques like decompression aware prefetching and QoS based bandwidth allocation could optimize workload management and maximize the benefits of PS5's unified architecture.

Next-generation architectures like the PS5 not just only upscale the user experience on the other hand, it also influence sustainability and society at large. The need for large DRAM pools has been reduced significantly by Sony after utilizing SSD streaming and decompression; it also lowers the manufacturing costs and energy consumption compared to simply scaling RAM capacity (Strickland, 2020; Patterson & Hennessy, 2020). These designs increase the lifespan of consumer hardware, allowing for higher fidelity without a proportional rise in cost, thus making gaming more accessible. However, mid-generation upgrades like the PS5 Pro (The Verge, 2024) reveal the conflict between innovation and sustainability. Additionally, frequent hardware refreshes contribute to electronic waste. This emphasizes the importance of energy-efficient design and software driven optimizations to increase console lifespans. Ultimately, the balance between performance growth and environmental responsibility will shape the success of modern architectures for both consumers and society.

Conclusion

To conclude, the PlayStation 5 shows how unified GDDR6 memory, a custom NVMe SSD, and hardware compression help achieve great performance while staying within power and cost limits.. Proposed optimizations such as a hybrid memory model, direct communication between the GPU and SSD, and turning off parts of the processor when not needed, could make the system faster and more efficient, but they would make the design more complicated. Overall, the PS5 highlights how the memory hierarchy directly influences user experience and emphasizes the importance of balancing performance, sustainability, and longevity in future systems.

References

- Copetti, R. (n.d.). *PlayStation architecture: A practical analysis*. <https://www.copetti.org/writings/consoles/playstation/>
- GeeksforGeeks. (n.d.). *Memory hierarchy design and its characteristics*. <https://www.geeksforgeeks.org/computer-organization-architecture/memory-hierarchy-design-and-its-characteristics/>
- Patterson, D. A., & Hennessy, J. L. (2020). *Computer organization and design: RISC-V edition* (2nd ed.). Morgan Kaufmann.
- Strickland, D. (2020). *Understanding the PS5's SSD: A deep dive into next-gen storage tech*. TweakTown. <https://www.tweaktown.com/news/71340/understanding-the-ps5s-ssd-deep-dive-into-next-gen-storage-tech/>
- The Verge. (2024, November 4). *PS5 Pro's actual specs revealed*. <https://www.theverge.com/2024/11/4/24287657/sony-ps5-pro-gpu-specs-unboxing-games-list-enhanced>